

Soham Tyagi

Brooklyn, New York | tysoham24@gmail.com

| [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

New York University

MS in Computer Engineering

Sept 2024 – May 2026

GPA: 3.7/4.0

Delhi Technological University

BTech in Electrical Engineering

Aug 2017 – June 2021

GPA: 9.2/10.0

EXPERIENCE

CinderellaAI

Co-founder

New York, USA

Jan 2026 – Present

- Building an **AI-powered fashion platform** featuring virtual try-on, a digital closet, outfit recommendation, and personalized styling. Selected for the **Leslie Labs Incubation Program**.

EG4 Electronics

New York, USA

Software Engineering Intern

Jan 2026 – June 2026

- Built and shipped an end-to-end solar PV forecasting pipeline: inverter telemetry from the EG4 API joined with Open-Meteo weather into a 43-feature set, feeding an **Optuna-tuned ensemble** (global XGBoost, per-device XGBoost, Random Forest) reaching **R^2 0.84** and **34.5% MAE improvement** over persistence baselines across **57 inverters**. Forecasts served through a **Flask API** backed by **RDS MySQL**.
- Designed the fleet's REST ingestion layer (pagination, multiple authentication domains, deduplication, location parsing) covering **3,630 GridBOSS inverters** across **3,569 sites** with idempotent, normalized MySQL loads.
- Ran fleet-wide analysis across **14,520 smart ports**, classifying AC-coupled, smart-load, and EV-charger usage and surfacing **210+ misconfigured units** to the operations team as actionable site-level reports.

Ernst & Young

New Delhi, India

Associate Consultant

Jan 2023 – Aug 2024

- Sole continuous technical resource for a global **pharmaceutical account**: CRM migrations, SAP BW integration, and production support.
- Migrated 24 Salesforce CRM objects from Amazon Redshift to Azure SQL DW as part of the client's program to retire a costlier legacy AWS stack, leading to **~68% cost reduction**; designed the SCD Type 2 stored procedures and cut a critical fact-table SP from **2+ hours to under 30 minutes** by converting CTEs to physical temp tables.
- Designed ADF pipelines for **76 of 117 SAP BW DSOs** using MDX (CROSSJOIN syntax, fiscal-year period variables), enabling a **daily Power BI sales dashboard** that replaced a weekly, week-lagged manual portal download.

Senior Analyst

Jan 2022 – Jan 2023

- Built the asset-health data platform for two Saudi national **utility clients**, ingesting IBM Maximo APM data via ADF (self-hosted IR), ADLS Gen2, and Databricks (PySpark) - **daily refreshes of 60+ Synapse tables** saving engineering teams **25–40 hours/month**.
- Co-designed the Synapse star schema and tuned distribution strategies, columnstore indexes, and materialized views - cutting a critical failure-cost query from **45+ minutes to under 8** and dashboard loads by **20–30%**; published **30+** governed KPIs in **Power BI**.
- Rebuilt failure-cost reporting as event-driven Python/SQL jobs on Azure Functions with Azure Monitor alerting - report turnaround from **~3 days to same-day**.

Analyst

Jul 2021 – Jan 2022

- Built the staging layer consolidating asset and outage data across **12 SEC operating regions** into Azure SQL via ADF, with SQL/Python cleansing cutting manual prep **~20%**.

PUBLICATIONS

Battery Charging Using Solar Photovoltaic System with Maximum Power Point Tracking

Jan 2022

Siddharth Samaddar, Ujjwal Rastogi, Soham Tyagi, Saurabh Mishra | [10.1007/978-981-16-6482-3_27](https://doi.org/10.1007/978-981-16-6482-3_27)

PROJECTS

Skua – Distributed Data Lakehouse

[GitHub](#)

Built a working distributed data warehouse from scratch: a **Raft-replicated** Iceberg-style table catalog, a Kafka-shaped replicated log with a Flink-shaped **exactly-once stream processor**, and a **vectorized SQL engine** with a hand-written parser, rule-based optimizer, and columnar storage format. Achieves **3–39× SQLite performance** on 5M-row analytical queries; chaos tests verify correctness under mid-query worker kills and Raft leader failures.

Ira – Hybrid Knowledge-Graph RAG

[GitHub](#)

Pure vector search doesn't understand relationships and pure knowledge graphs miss paraphrase. Ira is a retrieval system over a **Neo4j** graph of retail transactions that combines guardrailed **LLM-generated Cypher queries** with dense embedding search over product descriptions, fusing the two rankings via **Reciprocal Rank Fusion**. FastAPI service, interactive Cytoscape.js graph dashboard, docker-composed with local Neo4j and Ollama; hybrid retrieval outperforms either strategy alone on **precision@k** and **MRR** across a 30-query gold set, with latency dominated by **~1 s p50 LLM Cypher generation**.

Transit Pulse – Serverless Transit Ops Platform

[GitHub](#) | [Live Demo](#)

Live NYC subway operations dashboard implementing a full data platform on free static infrastructure: **GitHub Actions** cron ingests MTA **GTFS-Realtime** protobuf every 10 minutes, writes partitioned **Parquet** with a manifest, and enforces data-quality gates on feed freshness. Static **GitHub Pages** front-end runs **DuckDB-WASM** in the browser to query the Parquet directly via HTTP range requests.

SKILLS

Languages: Python, SQL/T-SQL, TypeScript, C/C++ , Java

Data Engineering: Kafka, Spark, Airflow, Dagster, dbt, Azure Data Factory, Snowflake, Databricks, dimensional modeling, medallion architecture

AI/ML: agentic workflows & AI orchestration, RAG, PyTorch, Tensorflow

Infra & DevOps: Docker, AWS, Azure, CI/CD (GitHub Actions), observability (CloudWatch, Azure Monitor)